DEIVA MANI KAILASAM

Senior AI Engineer | GenAI · Agentic AI · LLMOps

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SKILLS

- **GenAI & Agentic AI:** LLMs (Claude, GPT, Llama 3/3.2), LangGraph, LangChain, LlamaIndex, PydanticAI, Claude Agent SDK, OpenAI Agents SDK, MCP, multi-agent orchestration, Snowflake Cortex Analyst & semantic views, sqlglot, Hugging Face, PyTorch
- **RAG & Retrieval:** Hybrid search (BM25 + HNSW, RRF), cross-encoder reranking, BGE embeddings, tokenizer-aware chunking, Qdrant, MongoDB Atlas Vector Search
- **LLMOps, AgentOps & Evals:** DeepEval, Langfuse, OpenTelemetry, golden datasets, LLM-as-judge, faithfulness/hallucination evals, CI eval gates, LiteLLM, model routing, semantic caching, token/cost optimisation, fine-tuning, shadow/canary rollouts
- **AI Security & Governance:** OWASP LLM Top 10, prompt-injection defence, Bedrock Guardrails, PII/DLP redaction, entitlement-aware retrieval, human-in-the-loop controls, audit trails, citation-grounded answers (explainability), AI risk assessments, model/system cards, responsible AI
- **Languages & Data:** Python, TypeScript, Java, FastAPI, Spring Boot, Node.js, React, Next.js, Angular, PostgreSQL, MongoDB, Snowflake, DynamoDB, SQL Server, Redis (ElastiCache), Kafka (MSK), pandas, dbt, REST, SSE, gRPC, GraphQL

PROFILE

Senior software engineer with 9 years building cloud-native, multi-tenant platforms serving multiple clients and agencies, the last 3 taking GenAI and agentic AI from business problem to production. Works with business stakeholders to qualify use cases, define success metrics and build ROI cases, then designs and ships end to end: RAG pipelines, LangGraph multi-agent workflows, Cortex Analyst text-to-SQL agents, guardrails and evaluation suites on Amazon Bedrock. Full-stack across Python, TypeScript, Java/Spring Boot and React/Angular. Owns delivery from architecture to CI/CD and observability; mentored 5 engineers (3 promoted).

WORK EXPERIENCE

Kinesso, Omnicom Media Group AUG 2021 – JUL 2026
Senior Software Engineer (Full Stack, AI)

GenAI & Agentic Platforms (2023–2026)

- Led the pillar's GenAI initiative as technical lead of a 5-person team, owning the roadmap, architecture and delivery from use-case discovery to production.
- Partnered with business, product and analytics stakeholders to qualify GenAI use cases, define success metrics and build ROI cases; ran feasibility spikes on real data to drive go/no-go decisions before committing to build.
- Built a Snowflake analytics agent (LangGraph, Cortex Analyst) that replaced manual custom-dashboard builds in a legacy reporting system due for decommissioning, cutting creation time from ~15 min to under 1 min (p95).
- Raised text-to-SQL accuracy from ~60% to ~80% by building the semantic layer first: ran metric-definition workshops with data owners to settle conflicting KPIs (e.g. ROAS) as governed metrics in per-client semantic views; 90%+ before launch, improving further via golden-set growth and user feedback.
- Made agent output verifiable: generated SQL passes a sqlglot read-only guard and runs as the signed-in user; code-based critics check each chart spec against real columns and each narrative number against the result.
- Built shared sign-in and tenancy for agents and web apps used by multiple clients and internal agencies: Entra ID SSO, OAuth 2.0 On-Behalf-Of token exchange into Snowflake External OAuth, SCIM provisioning and per-client roles, so row-access and masking policies apply per user; data exposed to agents as tenant-scoped MCP tools.
- Designed multi-agent workflows in LangGraph and PydanticAI: Claude Sonnet on Bedrock for planning, reasoning and tool use, GPT via the OpenAI API, and intent routing that moved from Llama 3 on Bedrock to a fine-tuned Llama 3.2 1B classifier on SageMaker, behind a LiteLLM model gateway.
- Built multi-tenant RAG services (Python/FastAPI) serving 1K+ daily queries: self-hosted BGE embeddings (bge-large-en-v1.5) on SageMaker, tokenizer-sized chunks, BM25 + HNSW hybrid search fused with RRF (sub-200 ms p95 retrieval) and cross-encoder reranking over Qdrant and MongoDB Atlas, streaming cited answers over SSE.
- Cut LLM inference spend by ~40% through semantic caching on ElastiCache (Redis), prompt caching, cost-aware model routing and per-tenant token budgets in DynamoDB.

- **Cloud & Platform:** AWS (Bedrock, SageMaker, ECS/Fargate, EKS, Lambda, Step Functions, EventBridge, SQS/SNS, Kinesis Data Streams/Firehose, S3, RDS), Azure, Docker, Kubernetes (KEDA, Karpenter), Terraform, Helm, ArgoCD, GitHub Actions, Jenkins, Cosign, Datadog, CloudWatch, X-Ray, Grafana
- **Architecture & Delivery:** Stakeholder discovery & solution scoping, PoC-to-production delivery, system design, multi-tenant platforms & identity (Entra ID SSO, OAuth 2.0/OIDC, On-Behalf-Of token exchange, Snowflake External OAuth, SCIM), distributed & event-driven systems, microservices, CQRS/Saga, TDD (Pytest, Jest, Playwright, Zephyr), code review, mentoring, Agile

- Built event-driven RAG ingestion in which MSK and EventBridge events carrying S3 references trigger Step Functions and Lambda pipelines for chunking, embedding and tenant-scoped upserts, keeping answers current without batch re-indexing.
- Hardened RAG and agent workflows against OWASP LLM Top 10 risks with Bedrock Guardrails for PII/DLP, the fine-tuned Llama 3.2 1B classifier (Hugging Face, PyTorch) for prompt-injection and topic filtering, and entitlement-aware retrieval for tenant isolation.
- Established evaluation and observability: DeepEval suites on versioned golden datasets (LLM-as-judge, faithfulness and hallucination metrics) as CI release gates, Langfuse and OpenTelemetry tracing per agent step, and shadow/canary rollouts with human-in-the-loop approvals, kill switches, audit trails, AI risk assessments and model/system cards.
- Built LLM-based anomaly detection and a data-quality scoring engine, reducing undetected data-quality issues by ~75%.

Backend, Data & Streaming (2021–2026)

- Architected and led a multi-module Spring Boot 3 microservices platform (JWT/OAuth 2.0, OpenAPI) over PostgreSQL, SQL Server, Snowflake and MongoDB, with DB-, collection- and row-level tenant isolation and zero cross-tenant leakage across 12+ environments; tuned tail latency with async I/O, parallel queries and tiered caching (Caffeine, Redis).
- Built MSK (Kafka) pipelines streaming ad performance, spend and conversion events from Google Ads, Meta, Bing Ads and Adverity, with per-campaign ordering, idempotent upserts, jittered retries, replayable dead-letter queues and Kinesis Data Firehose landing raw events in S3.

Frontend & Product

- Built a 40+ component design system across React, Angular and Next.js, adopted by 6 internal products, with white-label theming for 5+ brands and AG Grid tables for self-serve analytics.

Platform, Quality & Leadership

- Owned CI/CD and infrastructure as code (Terraform, ArgoCD, Helm; EKS with KEDA and Karpenter autoscaling, ECS; Kyverno policies and Cosign-signed images; Jenkins and GitHub Actions with SonarQube gates), cutting merge-to-deploy time from hours to under 20 minutes.
- Cut MTTR by ~50% with CloudWatch, X-Ray, Datadog and Grafana dashboards and tracing; maintained 1,000+ unit and 100+ E2E tests at 80%+ coverage across Java, Python and UI layers.
- Mentored 5 engineers (3 promoted to mid-level), set coding and code-review standards across Python, TypeScript and Java, and introduced architecture decision records and incident runbooks so the team could operate and roll back systems independently.

OTHER NOTABLE PROJECTS

- **RAG Core Platform (PoC):** Config-driven RAG framework with 159 swappable components and 20+ reference pipelines, gated by CI evals; hybrid search outperformed dense-only retrieval on Recall@5 across a 120-query golden set. Secure by default: document-level ACLs, PII redaction, guardrails, a tamper-evident audit log and signed builds.
- **Agentic AI Core Platform (PoC):** Spec-driven agent framework with 60+ swappable components and 8 orchestration patterns, gated by trajectory evals on a 150-task suite. Secure by default: MCP tool permissions, human-in-the-loop approvals, budget caps, sandboxed execution, prompt-injection guardrails and a tamper-evident audit log.
- **AI-Assisted E2E Test Automation:** Built reusable Cursor skills that generate test scenarios, map them to Zephyr test cases and auto-create Playwright end-to-end tests, run on every pull request to block regression bugs before merge.

EDUCATION

- **Master of Information Technology**
  La Trobe University
  Jul 2019 – May 2021

- **Bachelor of Technology in Information Technology**
  Valliammai Engineering College
  Aug 2011 – May 2015

CERTIFICATIONS & AWARDS

- Google Cloud Certification on **GenAI** (2025)
- IBM Certification on **Prompt Engineering** (2025)
- **Best IT Project** Award — La Trobe University (2020)
- **Pursuit of Knowledge** Award, **On the Spot** Award, **Cornerstone** Award, **Star** Award — TCS (2016–2018)

ACE Contractors Pty Ltd DEC 2020 – JUL 2021
Software Developer (Full Stack)

- Worked directly with the company owner in weekly syncs to scope and prioritise high-ROI features, digitising timesheets and contracts and reducing payroll disputes.
- Architected and owned a progressive web app for construction operations on Azure (App Service, PostgreSQL Flexible Server, Functions, Blob Storage, AD B2C), serving 200+ daily active users across 10+ sites.
- Built a mobile-first Angular PWA with offline support (Service Workers, Cache API) for field workers, improving daily task completion by 40%, backed by Node.js/Express APIs with sub-100 ms p95 responses.
- Delivered CI/CD with Azure DevOps pipelines, Docker and Bicep, cutting deployment time from 2 hours to 12 minutes.

Tata Consultancy Services JAN 2016 – MAY 2019
Systems Engineer (Full-Stack Java/Angular)

- Led migration of a US telecom client's legacy Java EE monolith to a layered Spring Boot architecture, reducing technical debt and remediating critical security vulnerabilities.
- Migrated AngularJS 1.5 to Angular 5 incrementally (RxJS, lazy loading), achieving 3x faster initial page loads and a 65% smaller bundle.
- Re-platformed Oracle schemas to PostgreSQL with optimised indexing and stored procedures, cutting query times by 65%.
- Introduced test-first development (JUnit, Mockito, Selenium) to reach 85% unit-test coverage, and containerised services with Docker to support 8+ major releases consolidating legacy systems.